

NIGEL GEORGE

Data Analyst | Analytics Engineer | Python | SQL | Power BI

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PROFESSIONAL SUMMARY

Data professional with 5 years of Python and SQL backend engineering experience, now focused on data analytics and analytics engineering. Hands-on background in building production ETL pipelines, designing PostgreSQL star schema warehouses, and writing advanced SQL covering window functions, cohort analysis, fraud detection, and FP&A modeling. Graduate certificate in AI & Data Science from Loyalist College, Toronto (2022–2023). Canadian PR. Looking to bring strong engineering fundamentals and real analytical depth to a data analyst or analytics engineering team in Canada.

TECHNICAL SKILLS

Data & Analytics: SQL (Advanced), PostgreSQL, dbt, Star Schema Design, Dimensional Modeling, ETL/ELT Pipelines, Data Warehousing, Power BI, Data Cleaning, Cohort Analysis, Fraud Detection

Languages: Python (Pandas, NumPy, Flask, Django), Advanced SQL, JavaScript

Tools & Platforms: Git, GitHub, Docker, DBeaver, pgAdmin, VS Code, Postman, Linux

Methods: Data Modeling, Query Optimization, Agile, Technical Mentorship, Stakeholder Reporting

PROFESSIONAL EXPERIENCE

Freelance Data & Backend Engineer | Self-Employed | Toronto, ON *Feb 2024 – Present*

- Built concurrent Python ETL pipelines ingesting and transforming 250K+ records into PostgreSQL, achieving throughput of 8K–15K records per minute with a data completeness rate above 97%.
- Designed and optimized PostgreSQL schemas — including indexing strategies and schema redesign — reducing key query latency from ~250ms to 60–90ms, a 64–76% improvement.
- Developed a production web scraping and data ingestion framework using parallel workers, pagination controls, and retry logic, sustaining sub-3% failure rates across high-volume data collection runs.
- Implemented end-to-end data validation using pytest, achieving 90%+ code coverage across ingestion and transformation workflows to catch data quality issues before they reached downstream queries.
- Delivered analytical data APIs (Flask, JWT/Auth0, Pydantic validation) supporting 200–500 requests/minute in load testing, enabling reliable programmatic data access for client reporting pipelines.

Senior Backend Developer & Team Lead | Om Informatics | Mumbai, India *Jan 2020 – Apr 2022*

- Engineered complex PostgreSQL database structures — stored procedures, views, and query-tuned schemas — reducing analytical query execution times by 40% across core reporting workloads.
- Led 3 end-to-end data product development lifecycles from requirements gathering through production release, managing code reviews for a team of 3 developers.
- Designed and maintained application backends (Django, Flask) serving data to downstream reporting layers, reducing average API response latency by 60%.
- Managed production database maintenance, server configuration, and deployment pipelines in Linux environments, ensuring high availability for data-dependent business operations.

Software Developer | Hitesh Solutions | Mumbai, India *May 2017 – Jun 2018*

- Rebuilt relational database schemas in PHP and MySQL for a core client application, reducing production data incidents by 45% through improved data integrity constraints and normalization.
 - Built an Internal Messaging portal for the web app using PHP, JS, and HTML5.
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TECHNICAL PROJECTS

Canadian Financial Analytics Platform

GitHub: https://github.com/nigel-george/financial_sql_advanced

Python · PostgreSQL · Advanced SQL · Star Schema · Power BI · 2.3M rows

- Architected a production-grade financial data warehouse on PostgreSQL — designing the star schema (fact_transactions, dim_customer, dim_date, dim_category), ETL cleaning pipeline, and 8 business analytics modules covering fraud detection, customer lifetime value, cohort retention, FP&A variance, and Pareto revenue analysis across 30,000 simulated Canadian customers and 2.3M transactions.
- Fraud detection module flags velocity attacks and duplicate network payment payloads using LAG() OVER (PARTITION BY) with time interval deltas — identifying exact-match duplicate transactions (time_lapse = 00:00:00) hidden within 2.3M records.
- Applied advanced SQL throughout: window functions (LAG, DENSE_RANK, ROW_NUMBER, SUM OVER), rolling 3-month moving averages (ROWS BETWEEN), cohort retention matrices, dynamic CTE budget variance models, and running Pareto cumulative distributions.
- Implemented range partitioning by calendar year and targeted B-Tree indexing, achieving sub-2-second query response on 2.3M row scans that ran in 8–12 seconds on the raw unindexed table.

Yelp Analytics Engine

GitHub: <https://github.com/nigel-george/yelp-analytics-engine>

Python · PostgreSQL · dbt Core · Pydantic V2 · Docker · Star Schema · 7M records

- Built a Bronze → Silver → Gold medallion pipeline on PostgreSQL (Docker) processing 6.99M Yelp records — SHA-256 row hashing for idempotent ingestion, Pydantic V2 data contracts to quarantine malformed records, and declarative table partitioning by year that cut query times from ~1 second to milliseconds (verified with EXPLAIN ANALYZE).
- Gold layer modelled in dbt Core — LAG() window functions for monthly growth metrics, conditional aggregations for business performance trends, and a full test suite (unique, not_null, relationship) across all 7M review-to-business relationships. Staging layers materialized as Views; analytical marts as Tables for downstream BI performance.

Additional Projects: <https://github.com/nigel-george>

EDUCATION & CERTIFICATIONS

Ontario College Graduate Certificate — AI & Data Science | Loyalist College | Toronto, ON 2022 – 2023

- Capstone: Real-time object tracking pipeline using YOLOv7 — data preprocessing, model training with PyTorch, scikit-learn, and TensorFlow, and performance evaluation on custom video datasets.

Master of Science — Computer Science | SIES College, Mumbai University | Mumbai, India 2017 – 2019

- Graduated with Distinction — CGPA 8.1 / 10.
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INTERNSHIP & VOLUNTEERING

- SharePoint Developer Volunteer | Loyalist College, Toronto | Oct – Dec 2023 | Built an internal academic web module serving 1,000+ active students.
- Full Stack Developer Internship | Destiny Education & Travels, Mumbai | Jul 2017 – Apr 2019 | Completed my MSc CS Internship Program of 440Hrs. Created Customer Panel Module for their Web Application. Got hands-on experience with technologies like Python Flask, MongoDB, HTML5, CSS3, AJAX, and JavaScript.